A New DirectXion

A look at how Microsoft's DirectX 10 will change the nature of the graphics card

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The tech industry is in a constant state of déjà vu. Things come and go in cycles and yet constantly manage to surprise. Not too long ago, the central processor that powers much of the industry went through a pivotal change—it began life as a piece of silicon designed to work on a limited set of data, and expanded its role to a more encompassing one. At first it could only handle one application at a time, it could only address a limited set of memory at once, and you had to talk to it in its own "assembly" silicon-language. With time, the CPU's silicon budget increased, and with it, the number of tasks it could handle. Then you had time-sharing systems, wherein you were able to run multiple applications at the same time; moreover, each application was fooled into thinking that the entire system's memory was its own playground, thanks to virtualisation. Today, we take these changes for granted and their benefits seem mundane.

Interestingly, the graphics card of yore—that dedicated pixel-pushing slave of our entertainment age—is undergoing a very similar evolution. What was once a slab of silicon meant for very specific tasks is fast evolving into a vastly parallel silicon factory, populated by a transistor task force half-a-billion strong, and powering a multi-billion dollar gaming industry. Interestingly, the graphics chip is increasingly being referred to as the graphics processing unit (GPU) because much like the CPU, from which it derives the name, its work is becoming less specialised and more general.

Before we take a look at how this is happening, it is important to identify the three major forces that make up the industry. These are the independent hardware vendors (IHV) who design and produce the graphics chipsets and the graphics cards, the independent software vendors (ISVs or game developers) who design and produce the games, and last, the people whose job it is to talk to both the ISVs and the IHVs to create the software rules and directions which enable the graphics cards to talk to the games. This final piece of the puzzle forms the application programming interface (API). The three players are constantly involved in discussions on how to take the industry forward; these discussions are translated into the API, which then serves as the template upon which a graphics chipset is designed. Based on which the game can offer its eye-candy.

Here, we take a look at one such API—Microsoft's upcoming DirectX 10 (DX10), more specifically the Direct3D element of DX10. Why was this step taken, and where it will lead us?

Why DirectX 10?

With the introduction of NVIDIA's very first GeForce chipset, the graphics industry took its first tentative step towards the GPU. The path that was taken then, today leads to a place where the graphics chip does more than push pixels. This semblance to a CPU is because of the identified need for the graphics processor to gain independence from the CPU. With DX10, engineers hope to break the shackles that tie graphics processing to the CPU, not only speeding up rendering but also granting the GPU more power to push floating point and integer data—power that will enable it to accelerate sundry tasks from 3D graphics, to physics, to audio, to DVD playback.

DirectX 10 is thus a very important step in taking the GPU forward into unknown regions. The API is primarily designed to interface with a more generic graphics processor. With the power of DX10 and the hardware supporting it, games of tomorrow will be richer, faster, more interactive, and more detailed. DX10 will take the gaming industry another step closer to the holy grail of truly realistic visuals.

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